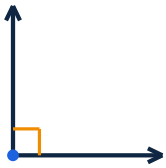
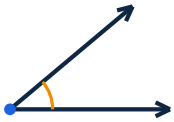
Angles

An angle is a turn between two rays. We measure it in degrees.

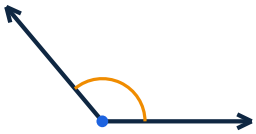
A full turn is 360 degrees. A square corner is 90.



right: 90°

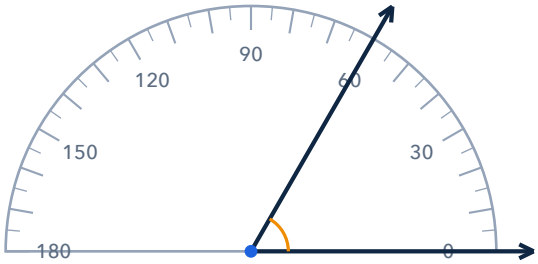


acute: under 90°



obtuse: over 90°

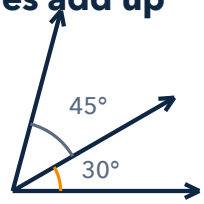
Read a protractor



1. Put the dot on the vertex.
2. Line up one ray with 0.
3. Read where the other ray crosses.

60 degrees

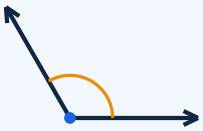
Angles add up



30 + 45 = ☐ degrees

Two angles side by side: add their degrees.

Your turn!

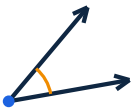


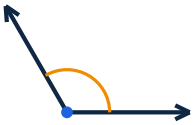
acute right obtuse

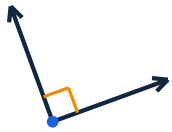
Angles: Practice

Square corner = 90 degrees. Straight line = 180 degrees.

Part 1 Acute, right or obtuse? Compare with a square corner.

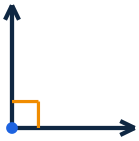
1)  acute right obtuse

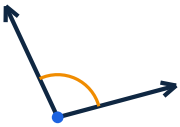
2)  acute right obtuse

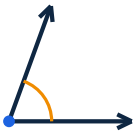
3)  acute right obtuse

4)  acute right obtuse

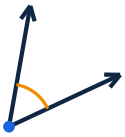
5)  acute right obtuse

6)  acute right obtuse

7)  acute right obtuse

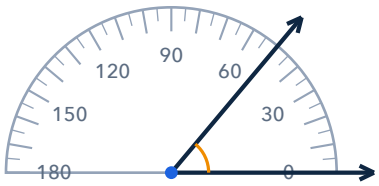
8)  acute right obtuse

9)  acute right obtuse

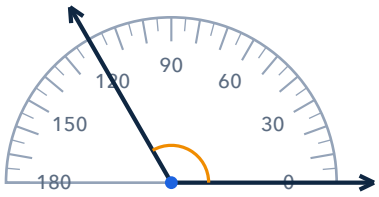
10)  acute right obtuse

Angles: Practice (continued)

Part 2 Read the protractor How many degrees?

11) 

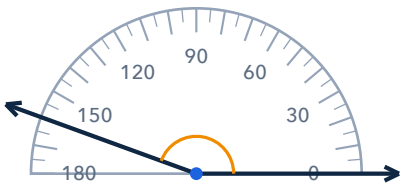
☐ degrees

12) 

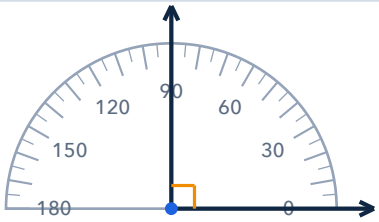
☐ degrees

13) 

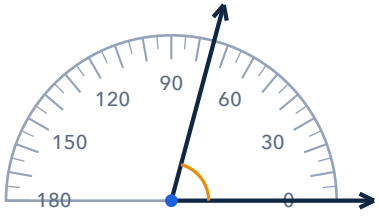
☐ degrees

14) 

☐ degrees

15) 

☐ degrees

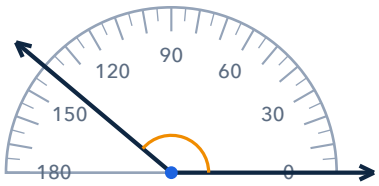
16) 

☐ degrees

Angles: Practice (continued)

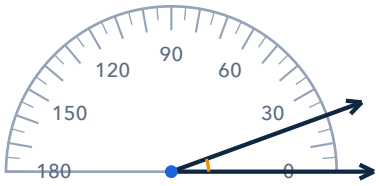
Part 2 Read the protractor (continued)

17)



☐ degrees

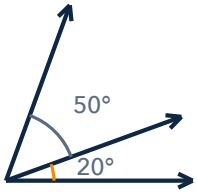
18)



☐ degrees

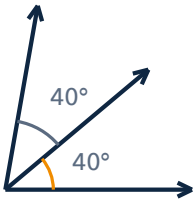
Part 3 Angles add up Add the two angles.

19)



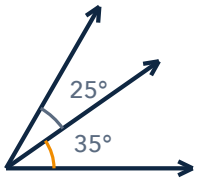
20 + 50 = ☐ degrees

20)



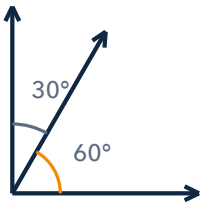
40 + 40 = ☐ degrees

21)



35 + 25 = ☐ degrees

22)



60 + 30 = ☐ degrees

Part 4 Find the missing angle No help this time. Together they make the angle shown.

23) 35 + ☐ = 90 degrees

24) 120 + ☐ = 180 degrees

25) 45 + ☐ = 90 degrees

26) 70 + ☐ = 180 degrees

27) 25 + ☐ = 90 degrees

28) 150 + ☐ = 180 degrees

Angles: Practice (continued)

Part 4 Find the missing angle (continued)

29) 60 + ☐ = 90 degrees

30) 95 + ☐ = 180 degrees

Part 5 Add angles No help this time.

31) 45 + 30 = ☐ degrees

32) 60 + 25 = ☐ degrees

33) 90 + 45 = ☐ degrees

34) 15 + 75 = ☐ degrees

35) 100 + 50 = ☐ degrees

36) 35 + 40 = ☐ degrees

37) 70 + 80 = ☐ degrees

38) 125 + 30 = ☐ degrees

Angles: Practice (continued)

Part 6 Turns No help this time. A full turn is 360 degrees.

39) 1/4 turn = ☐ degrees

40) 1/2 turn = ☐ degrees

41) 3/4 turn = ☐ degrees

42) 1 full turn = ☐ degrees

43) 2 right angles = ☐ degrees

44) 3 right angles = ☐ degrees

Part 7 Make a right angle

45) 67 + ☐ = 90 degrees

46) 43 + ☐ = 90 degrees

47) 17 + ☐ = 90 degrees

48) 58 + ☐ = 90 degrees

49) 83 + ☐ = 90 degrees

50) 32 + ☐ = 90 degrees

Angles: Practice (continued)

Part 7 Make a right angle (continued)

51) 15 + ☐ = 90 degrees

52) 46 + ☐ = 90 degrees

53) 16 + ☐ = 90 degrees

54) 6 + ☐ = 90 degrees

55) 61 + ☐ = 90 degrees

56) 52 + ☐ = 90 degrees

57) 50 + ☐ = 90 degrees

58) 38 + ☐ = 90 degrees

59) 47 + ☐ = 90 degrees

60) 22 + ☐ = 90 degrees

61) 33 + ☐ = 90 degrees

62) 85 + ☐ = 90 degrees

Angles: Practice (continued)

Part 8 Make a straight angle

63) 86 + ☐ = 180 degrees

64) 124 + ☐ = 180 degrees

65) 103 + ☐ = 180 degrees

66) 132 + ☐ = 180 degrees

67) 159 + ☐ = 180 degrees

68) 139 + ☐ = 180 degrees

69) 160 + ☐ = 180 degrees

70) 134 + ☐ = 180 degrees

71) 137 + ☐ = 180 degrees

72) 122 + ☐ = 180 degrees

73) 77 + ☐ = 180 degrees

74) 114 + ☐ = 180 degrees

Angles: Practice (continued)

Part 8 Make a straight angle (continued)

75) $48 + \square = 180$ degrees

76) $144 + \square = 180$ degrees

77) $154 + \square = 180$ degrees

78) $106 + \square = 180$ degrees

79) $43 + \square = 180$ degrees

80) $152 + \square = 180$ degrees

Part 9 Remember? Problems from earlier packets.

81) area: $16 \times 9 = \square$ sq ft

82) 6 m $= \square$ cm

83) $1\frac{3}{8} + \frac{6}{8} = \frac{\square}{8}$

84) $0.56 \square 0.6$

85) $\frac{9}{10} = \square$

86) $1{,}206 \div 3 = \square$

Angles: Practice (continued)

Part 10 Word problems Draw the angles.

87) A door opens 30 degrees, then 45 degrees more. How far has it opened?

Number sentence _____

Answer ☐ degrees

Draw or show your work

88) The hands of a clock make a square corner at 3:00. How many degrees is that?

Number sentence _____

Answer ☐ degrees

Draw or show your work

89) A skater turns half a circle. How many degrees does she turn?

Number sentence _____

Answer ☐ degrees

Draw or show your work

90) A pizza slice has a 60 degree corner. How many slices fit around the whole pizza?

Number sentence _____

Answer ☐ slices

Draw or show your work

91) Challenge: Two angles make a straight line. One is 115 degrees. How big is the other?

Number sentence _____

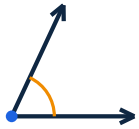
Answer ☐ degrees

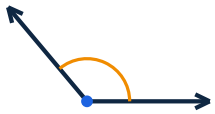
Draw or show your work

Check-In: Angles

Show what you know. No help this time. Score: _____ / 8

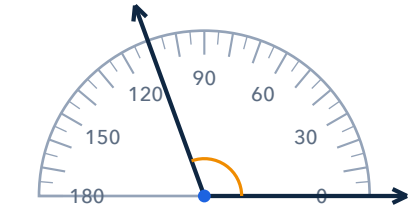
Part 1 Acute, right or obtuse?

1)  acute right obtuse

2)  acute right obtuse

Part 2 Read the protractor

3)  _____ degrees

4)  _____ degrees

Part 3 Find the missing angle

5) 65 + ☐ = 90 degrees

6) 75 + ☐ = 180 degrees

7) 55 + ☐ = 90 degrees

8) 55 + ☐ = 180 degrees